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Innovative Thixotropic and Reactive Cement Design to Cure Lost Circulation

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Abstract

The challenges involved with using remedial cementing to cure lost circulation, either in the open hole or behind casing to restore zonal isolation and well integrity are well known. Curing lost circulation has become more difficult, risky, costly and complicated in wells with lithology that contain large natural fractures and cavernous formations. Losses can be both static and dynamic in the range of 100 to 300 bbls per hour and can cause long delays in drilling and completion operations or even lead to uncemented sections that jeopardize zonal isolation. This manuscript focuses on a specific multi-stage reactive and thixotropic train of fluids, adopted from similar lost circulation treatments performed in the past and known to be effective.

A train of viscous reactive pills, which consisted of Calcium Chloride brine and Sodium Silicate solution, were pumped ahead of and between each of three progressively thixotropic and aggressive cementing fluids. The train of cementing fluids was comprised of low-density cement with strong thixotropic properties reducing the hydrostatic head and rapidly gelling up against the loss zone when contacting the reactive pills followed by nominal density cement with a stronger degree of thixotropy followed by a normal cement slurry with minimal thickening time. The intermixing of these fluids forms a very rapid gel structure in the lost circulation zone, becoming progressively more and more immobile until the loss zone is eventually filled.

The fluid train was deployed and has been in use to cure many lost circulation situations for many different wells. Prior to this deployment, the normal practice was to spot a balanced cement plug in the loss zone, which usually required multiple attempts and delayed operations several weeks and many times the lost circulation remained unchanged. The new method reduced the number of attempts to cure lost circulation and sometimes cured lost circulation on the first attempt. This has reduced the number of sections that require drilling blind and also reduced the amount of uncemented sections behind casing that can lead to corrosion, cross flow and other loss of well integrity.

The novelty of the procedure is the combination of many reactive fluids and thixotropic cement slurries instead of just one reactive fluid and one cement slurry and this new method is more effective than the old method.

Introduction

Lost circulation events can occur at any depth and in any hole size and also lost circulation can range from 10 to 400 barrels per hour or even complete losses. An operator may encounter losses while circulating, typically called dynamic losses, or even when the well is static, typically called static losses. The situation can become worse if the loss zone is encountered below any intervals with flow zones that require a minimum amount of hydrostatic pressure to prevent influx into the wellbore, forcing the operator to pump kill mud down the annulus until the losses are cured. Several days or even weeks of delays and non-productive time can accumulate after encountering lost circulation.

There are many methods available to cure losses, some of which include pumping cement. This paper will examine how a traditional method of pumping cement to cure losses was completely overhauled to become more aggressive, effective and efficient. The overhaul involved pumping fluids, highly reactive with cement, such as calcium chloride brine and sodium silicate solution. These techniques date back to the 1960's and 1970's and proved themselves still relevant today. This paper will also explain how the placement method was changed from a balanced plug concept to a squeeze concept, which was a paradigm shift in the use of cement as a lost circulation treatment.

Statement of Theory and Definitions

Lost circulation is defined as the unintentional loss of drilling fluid from the wellbore into the surrounding formation during drilling operations. This occurs when the fluid pressure exceeds the formation's fracture pressure or when the fluid flows into highly permeable zones, natural fractures, or cavities. Lost circulation can lead to reduced drilling efficiency, formation damage, well control issues, and increased operational costs. It is a critical challenge in drilling, often requiring mitigation techniques such as using lost circulation materials (LCM), adjusting mud properties, or employing specialized drilling practices. Unfortunately, without specialized downhole tools or drilling through a section with known lithology, it is very difficult to know what is causing the lost circulation. For most lost circulation situations, all that is really known is the rate of losses and the depth, at which it occurred, so the best treatment for curing lost circulation can be difficult to determine (Canson, 1985; Wang, 2008; Metcalf, 2011).

Most operators have developed a decision tree, based on offset wells and previous experience, to help make the best choices in materials and methods to cure losses and what to do next when losses are still not cured. At some point the operator will decide to try to cure losses by pumping cement slurry into the loss zone. The idea behind using cement slurry to cure losses is that the cement slurry will fill the fractures, voids, vugs, caves, or faults and when the cement hardens it will seal and block the flow path. As opposed to drilling fluid treatments with LCM, which are designed to block the entrance to the void space with fibers or particulates or a combination of both. Or in the case of natural or induced fractures, drilling fluid treatments with LCM can also be designed to strengthen the near well bore lithology. For partial to moderate losses, spotting a balanced cement plug across the depth of the loss zone is often successful or only requires two attempts. However, for severe and total losses, spotting a balanced cement plug tends to have little to no effect on the losses.

A balanced plug is a cement plug placement technique where cement slurry is pumped through a work string (drill pipe or tubing) to a specific depth in the wellbore, with the volumes of cement, spacer and displacement fluid calculated to achieve hydrostatic equilibrium between the fluid columns inside the work string and in the annulus. The balance ensures that the cement slurry remains at the desired depth after placement, preventing it from moving up or down due to pressure differences (Nelson and Guillot, 2006). The method includes a pumping sequence of a spacer or chemical wash, which is pumped ahead of the cement slurry to minimize contamination with drilling fluid, followed by the cement slurry, and then a spacer and displacement fluid behind it. The volumes are designed so that the heights of the fluids (cement, spacer, and mud) are equal inside the pipe and in the annulus at the end of displacement. The method also involves

hydrostatic balance, which aims to equalize hydrostatic pressures at the base of the cement plug, ensuring the cement stays in place during setting. This is achieved by carefully calculating the volumes based on wellbore geometry, pipe dimensions, and fluid densities (Nelson and Guillot, 2006). After the balanced plug technique is performed, the work string must be pulled out of the cement to a depth above the theoretical top of cement where the operator can circulate or reverse circulate to clean any traces of cement out of the work string. Typically, the length of a balanced cement plug is limited to 500 ft due to the time required to pull the work string out of the cement plug. Based on the description of the balanced plug method, it becomes clear that this method may not be appropriate for curing total or severe losses with cement.

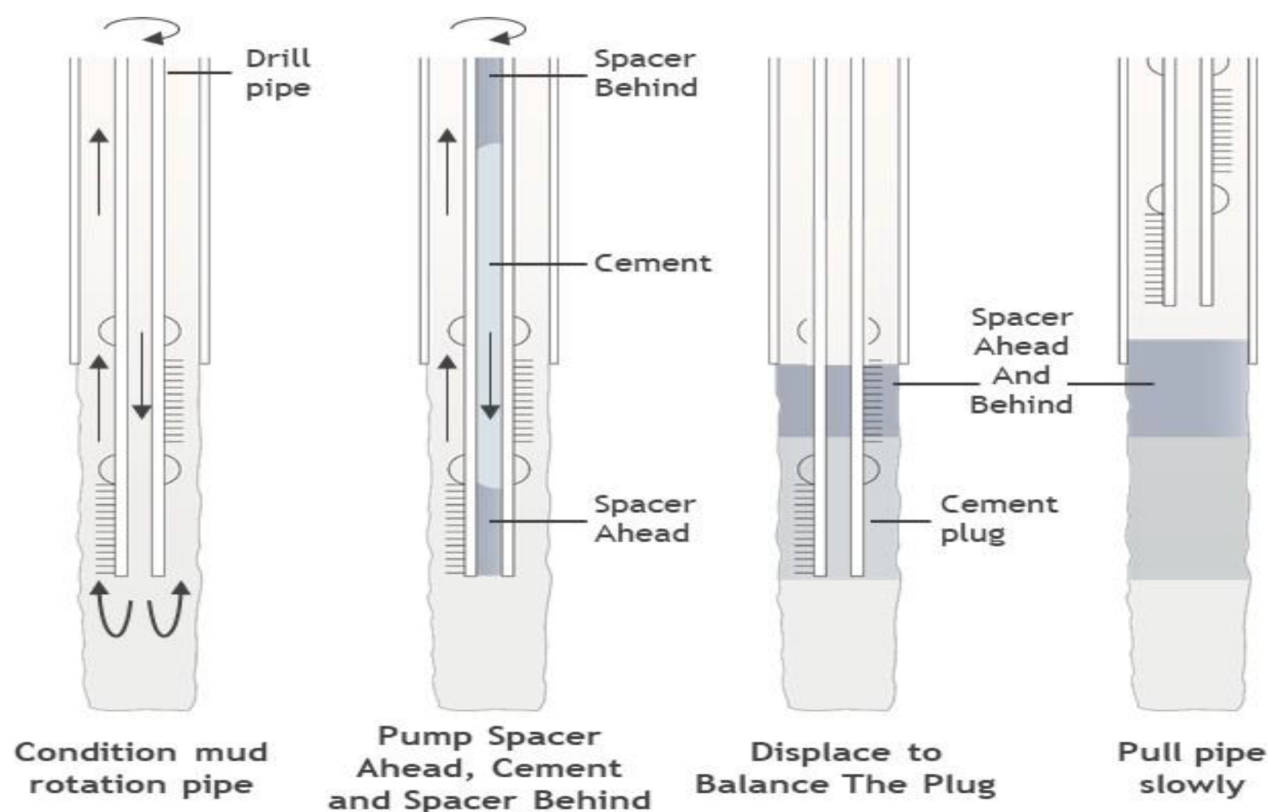
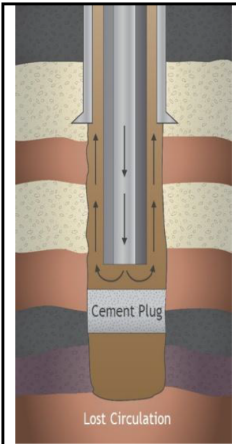


Figure 1—Process for Spotting A Balanced Cement Plug in The Wellbore

Although spotting cement in the wellbore with the balanced plug method is a dependable and accurate method, the volume of cement is limited based on the length of the plug. When trying to cure total or severe losses with cement, it will most likely require very large volumes of cement. See Table 1 below, which demonstrates how much time a 500 ft cement plug will remain in the wellbore for various hole sizes and various loss circulation rates. If the loss rate is 400 bbls/hr then the cement will only remain in the wellbore one hour for a large hole size (like a conductor section) and only 3 minutes for a small hole size (like a production section). Unless the cement can set in less than an hour, or for a small hole size less than 3 minutes, the cement slurry will be consumed by the lost circulation zone before it can seal it off. Often, many repeated attempts are required to cure losses using this technique, and many times this technique does not cure the losses, even after 10 or more attempts. To make this method more effective, something needs to be done to cause the cement to rapidly set or develop a rigid gel structure or to give the cement more time to set before it is consumed by the loss zone.

Table 1—Residence Time of a 500 ft Cement Plug in Various Hole Sizes vs Various Loss Rates


Hole Size	500 ft Plug Cement Volume	Residence Time with 400 BBL/HR Loss Rate	Residence Time with 300 BBL/HR Loss Rate	Residence Time with 200 BBL/HR Loss Rate	Residence Time with 100 BBL/HR Loss Rate	Residence Time with 50 BBL/HR Loss Rate	Residence Time with 25 BBL/HR Loss Rate
A	381 BBLS	57 MIN	76 MIN	114 MIN	228 MIN	457 MIN	914 MIN
B	235 BBLS	35 MIN	47 MIN	71 MIN	141 MIN	282 MIN	564 MIN
C	124 BBLS	19 MIN	25 MIN	37 MIN	75 MIN	149 MIN	298 MIN
D	70 BBLS	10 MIN	14 MIN	21 MIN	42 MIN	84 MIN	168 MIN
E	34 BBLS	5 MIN	7 MIN	10 MIN	20 MIN	41 MIN	82 MIN
F	17 BBLS	3 MIN	3 MIN	5 MIN	10 MIN	20 MIN	40 MIN

Cement slurries designed to be placed in the wellbore using the balanced plug method will have a thickening time in the range of 3 to 5 hours or more depending on the depth. Even though the placement time required may only be 1 to 2 hours, the time required to pull the work string out of the cement and circulate must be included in the thickening time. Otherwise the work string may become stuck in cement before it is pulled out above the cement plug. As a result of this amount of thickening time the subsequent time required for the cement slurry to transition from liquid to solid will also be extended. Again, according to [table 1](#) the slurry needs to set in less than an hour in order for it to be an effective treatment for lost circulation. There are chemicals that can be used to accelerate the setting time of a cement slurry, which can be added to the slurry mix fluid, or blended into the bulk cement blend, or post added to the slurry after it is already mixed.

Sodium silicate is a commonly used inorganic gel. Sodium silicate, which is normally used as a cement extender; however, it also has an accelerating effect. Sodium silicate reacts with Ca^{2+} ions in the aqueous phase of the cement slurry to form additional calcium silicate hydrate (C-S-H) phase nuclei, thus hastening the end of the induction period ([Nelson and Guillot, 2006](#)). Sodium silicate can be in solid form, called sodium metasilicate (SMS), Na_2SiO_3 . SMS can be dry blended in the bulk cement blend or pre-hydrated in the mix fluid before mixing. The recommended concentration range of SMS is 0.2% to 3.0% by weight of cement (BWOC). Sodium silicate can also be in liquid form, called water glass or liquid glass, $\text{Na}_2\text{O} \cdot (3-5) \text{SiO}_2$. Liquid sodium silicate is added to the mix fluid before mixing cement and the recommended concentration range is 0.2 to 0.6 gallons per sack of cement (GPS). When used in the recommended ranges, either SMS or liquid sodium silicate provides controllable acceleration of the cement slurry and also give the cement slurry thixotropic properties. When liquid sodium silicate is added to cement in concentrations far above the recommended range it will cause the cement to set in less than 5 minutes. See [figure 2](#) where a 50:50 solution of water and liquid sodium silicate was added to a 15.8 pound per gallon (PPG) cement slurry in a laboratory setting. The combination of cement slurry and sodium silicate solution became unmixable in less than one minute.



Figure 2—Mixing 15.8 ppg Cement Slurry with 50:50 Solution of Water and Sodium Silicate

Sodium silicate can be externally catalyzed. It reacts with brine to form a stiff gel in formation pores wherever they contact brine (Kabir, 2001). When sodium silicate and calcium chloride come into contact, the calcium chloride reacts with the silicate to precipitate calcium silicate ‘slush’ and the forming of sodium chloride in solution. The chemical reaction is near instantaneous and the resulting solution produces a thick fluid containing glass like flakes forming a slush, which is precipitated calcium silicate. A simple mixing test of these two fluids produces an interesting demonstration of the rapid reaction (Olsen et al., 2019). See figure 3, where a 50:50 solution of water and sodium silicate was added to a milling fluid saturated with calcium chloride in a laboratory setting. The combination of the two fluids became unmixable in less than one minute, resulting in a slush of glass like flakes of precipitated calcium silicate. In order to get the calcium chloride to precipitate and react with all of the sodium silicate solution. Calcium chloride brine is typically close to the saturation point. Studies have shown that at least 1.5% calcium chloride by weight is needed to fully react with all of the sodium silicate.



Figure 3—Mixing of Saturated CaCl Fluid and 50:50 Solution of Water and Sodium Silicate

Thixotropy is a term used to describe the property exhibited by a system that is fluid under shear (i.e. pumping or agitation) but develops a gel structure when the shear is stopped. In practical terms, thixotropic systems are fluid during mixing and displacement but rapidly form a rigid, self-supporting gel

structure when pumping ceases. When a thixotropic slurry enters a lost circulation zone, the velocity of the leading edge decreases and a gel structure starts to form. As the gel strength develops, resistance to flow increases until the entire zone is plugged. The self-supporting property of thixotropic cements is useful across formations with low fracture gradients. When ordinary slurries pass over a weak zone, the increase in hydrostatic pressure can cause formation breakdown. As a result, the top of cement falls back because some of the hydrostatic pressure is transmitted to the formation face and casing walls. As a rule, thixotropic slurries behave as Bingham plastic fluids under stress; consequently, their behavior is defined by a yield value (τ_y) and a plastic viscosity (μ_p). τ_y is a theoretical value concerning the behavior of a fluid under conditions of shear. With thixotropic slurries, τ_y would be the shear stress necessary to initiate movement, i.e., measured at zero shear rate. For a nonthixotropic fluid, the yield value remains the same whether the shear rate is increasing or decreasing. There is no change in the physical structure of the fluid during the static period, and the pressure required to move the fluid does not change with time. In the case of a thixotropic fluid, the yield point is exhibited only upon the withdrawal of shear stress. If there is a lapse of time after the withdrawal of shear, a force greater than the yield point will be required to move the fluid, see figure 4. The difference between the "gel strength" and the yield point gives a measure of the "degree of thixotropy" of the fluid. This can be measured in the lab by performing the 10 second, 10 minute and 30-minute gel strength test after measuring the rheology of the fluid on a rotational viscometer (Nelson and Guillot, 2006).

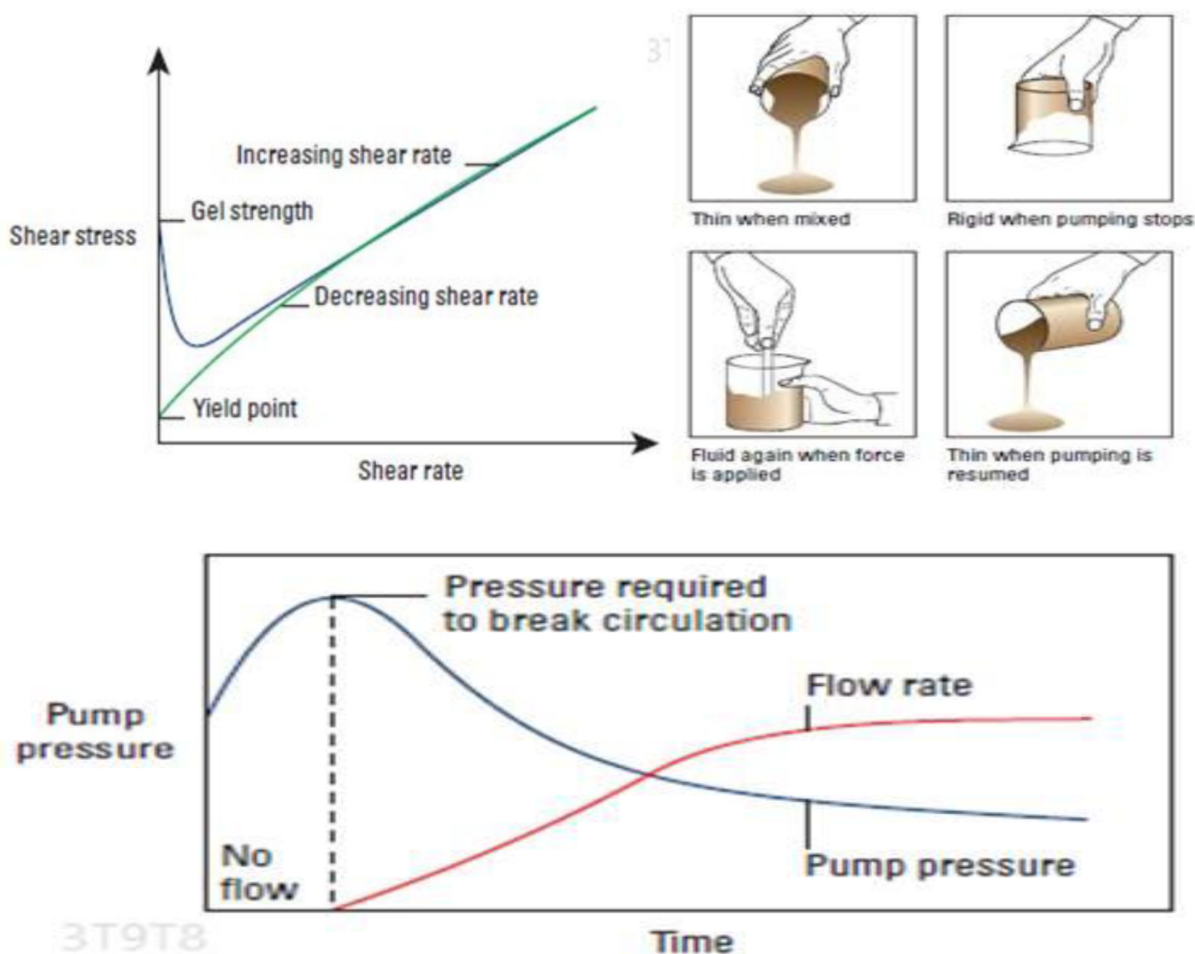


Figure 4—Examples of Fluids with Thixotropic Properties. Gel strength develops rapidly when shear rate is removed. Gel strength is strong enough for the fluid to become self-supporting. When a thixotropic fluid becomes static, it takes higher pump pressure to break circulation. With enough shear stress the gel can be broken and the fluid goes back to original viscosity (Nelson and Guillot, 2006).

A cement squeeze refers to a remedial cementing operation where cement slurry is pumped under pressure into a wellbore to seal off voids, perforations, repair casing leaks, or isolate specific zones to prevent unwanted fluid flow. The process involves pumping cement through drill pipe or tubing, which is placed near a target area and forcing cement into that target area under pressure. A cement squeeze is a common technique used to cure lost circulation in drilling operations. A cement squeeze is employed to seal off zones causing lost circulation by injecting cement slurry under pressure into the loss zones. Cement slurry is pumped through the drill string or tubing to the lost circulation zone, either through perforations if the loss zone is behind casing, or directly into the open hole. The mechanism behind forcing the cement into the loss zone can be achieved by either shutting in the well using the blow out preventer (BOP) or by running a downhole tool, such as a retrievable or drillable packer. See [figure 5](#).

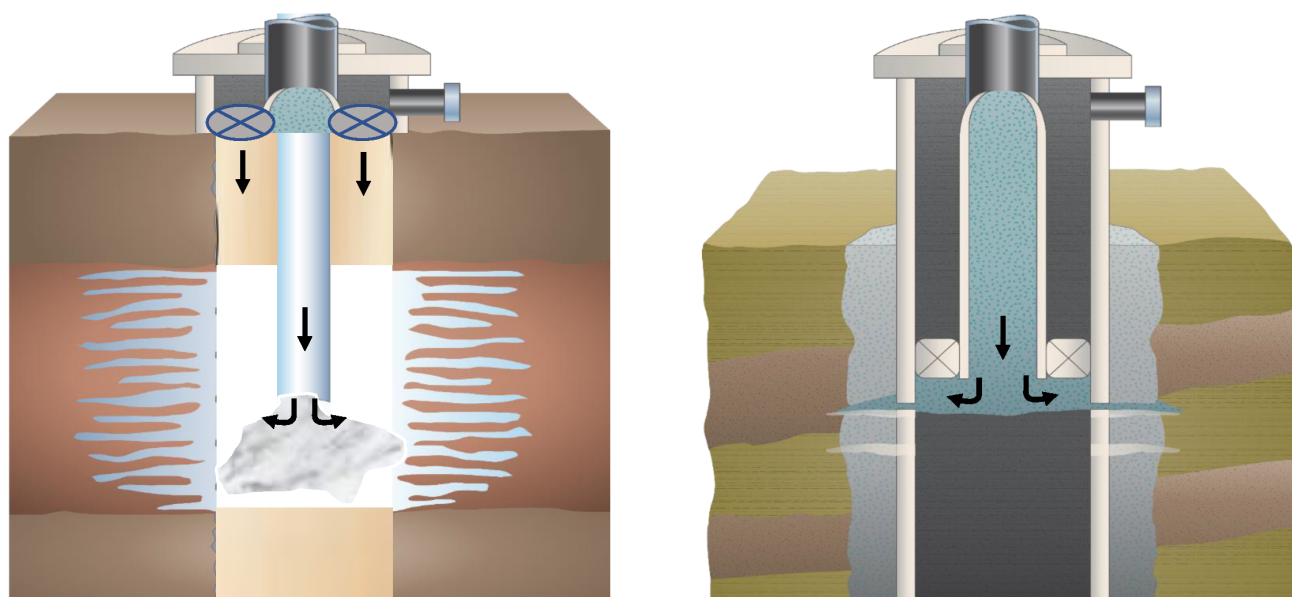


Figure 5—Illustration of a cement squeeze using the BOP method and using a downhole packer

In the case of using the BOP for a cement squeeze, the operator runs into the well with the work string to the depth of the loss zone. Then the annular preventer or the pipe rams are closed around the work string to prevent returns to surface. The spacer and cement is then pumped into the work string down to the loss zone where it is forced into the loss zone under pressure. In the case of severe or total losses the hydrostatic pressure of the spacer and cement is enough to force the fluids into the loss zone, so closing the BOP may not be necessary. To prevent the pipe from getting stuck by any cement that may find its way around the drill pipe during the operation, the work string is reciprocated so any over pull can be detected. Also, mud can be pumped down the back side of the work string using the choke and kill line of the BOP, which will also help flush any cement that tries to creep up around the work string during the squeeze. After the squeeze operation the BOP is opened and the work string is picked up to a safe depth and circulated clean. At this point circulation pressure is noted and compared to the circulation pressure prior to pumping the cement squeeze, to determine if any reduction in losses were achieved. During the cement squeeze, if there were any signs of over pull, increase in pressure, or change in hook load, then mixing and pumping cement would be shut down, the BOP would be opened to pull the work string clear of cement and displace cement out of the work string. The advantage of this method over the balanced plug method is that the volume of cement is no longer limited to a theoretical length of a cement plug. Cement can be mixed and pumped continuously until there are signs of the loss zone being sealed. Another advantage is that the thickening time of the cement can be reduced to only the time required to be pumped down the work string and into the loss zone.

In the case of using a downhole tool for a cement squeeze, the operator runs and sets a packer into the well to a depth above perforations, (for squeezing behind the casing), or in the previous shoe (for squeezing the open hole below the previous shoe). The work string is connected to the downhole packer in such a way that fluids can be pumped down the work string and forced below the packer under pressure. Spacer and cement are pumped down the work string and forced into the voids behind the casing or forced into the open hole and into the loss zone. Since the annulus above the downhole tool is sealed by the packer there is no risk of cement sticking around the casing. After the squeeze the work string is stung out of the packer and circulated clean. After waiting on cement the operator can try to sting back into the packer to establish an injection rate or pressure test to see if the losses have been reduced, or repeat the squeeze again. The operator can also choose to drill out the packer and try to establish circulation and test to see if the losses have been reduced or cured. This squeeze technique has the same advantage as the BOP method, as it allows large volumes of cement to be pumped below the packer without risk of getting stuck. Like the BOP method, cement can be mixed and pumped continuously until there are signs of the loss zone being sealed. And also, like the BOP method, the thickening time of the cement can be reduced to only the time required to be pumped down the work string and below the packer.

Putting all the pieces together, it is possible to accurately place cement slurry into a loss zone, without restrictions on cement slurry volume and at the same time there is a way to force the cement to rapidly set before it is consumed by the loss zone. Instead of trying to spot a balanced cement plug across a loss circulation zone with a small volume of cement slurry that is designed to have a long thickening time and thus a long time to set and develop strength, a large volume of cement slurry designed to set rapidly and with thixotropic properties can be pumped into the loss zone as a cement squeeze. Also, fluids that react with cement like liquid sodium silicate and calcium chloride can be pumped ahead of and behind the cement slurry to force extremely rapid setting.

Description and Application of Equipment and Processes

Severe lost circulation or even total losses, either static or dynamic, can now be treated with in a completely different fashion than the balanced plug method. The novel treatment for severe to total losses is a sequence of fluids, a fluid train, consisting of calcium chloride brine, sodium silicate solution and cement slurry with a small amount of water or spacer between each fluid to prevent any of the fluids mixing together inside the work string, which would result in plugging the work string. The idea behind this method is that these fluids will come into contact with each other in the loss zone and react rapidly to cure the losses (Olsen et al., 2019). The effectiveness of use of this fluid train has been documented for curing losses (Canson, 1985). The cement slurry used for this novel treatment is a thixotropic cement slurry designed to have a lower density in the range of 11.5 PPG to 14.5 PPG. The thixotropic properties of the cement slurry allow it to have a rapid gel strength development and viscous rheology profile. The low density of the cement slurry reduces hydrostatic pressure and also prevents the compressive strength from becoming high enough to cause unintended sidetracking when drilling out excess cement. Although the strength of the cement is designed to be lower than conventional cement, it is designed to develop very rapidly (Olsen et al., 2019).

The combination of these fluids provides two compounding benefits. First, the reaction between sodium silicate and calcium chloride is near instantaneous, so it has a better chance creating blockage inside the loss zone when trying to cure severe lost circulation. The precipitated calcium silicate flakes can then provide enough of a restriction to the fluid flow in the loss zone to allow the thixotropic cement to gel and set up before it is consumed by the loss zone. Second, the cement slurry gel strength development will accelerate when it contacts the sodium silicate and calcium chloride mixture (Olsen et al., 2019). Both of these chemicals individually are accelerants of cement (Smith, 1974).

The volumes and concentration of calcium chloride brine, sodium silicate solution, and cement slurry are determined by hole size, loss rate, work string volume and experience from offset well data or previous

studies. In the study of (Garba et al., 2014) the conclusion was that the concentration of sodium silicate has little effect as the cement and sodium silicate hardens at all concentrations. Also, they concluded that the volume of fresh water or spacer between the cement and sodium silicate has little effect. What had the most detrimental effect was contamination with brine, oil-based mud (OBM) and formation fluids. This suggests that the volume of cement needs to be large enough to reduce the contamination with fluids in the well bore.

The fluid volumes for the lost circulation treatment discussed in this paper were based on work string volume, volume of the hole with excess and the rate of losses. The volume of the calcium chloride brine and sodium silicate solution was chosen to be 100 bbls based on batch mixer volumes. The volume of fresh water or spacer between the fluids was chosen to be 250 ft of work string, which is about 4 to 5 bbls depending on work string pipe size. The volume of cement was chosen to be an equivalent amount to achieve at least one-hour residence time in the open based on the rate of the losses. For example, if the loss rate is 400 bbls per hour, then the cement volume should be a minimum of 400 bbls.

The thixotropic cement slurry is designed to develop rapid gel strength. The degree of thixotropy is measured in the lab using the rotational viscometer and also the consistometer. After measuring the rheology in the rotational viscometer, the 10 second and 10-minute and 30-minute gel strength is measured. The 10-minute gel strength should be at least twice the 10 second gel strength. The 30-minute gel should be at least four times the 10 second gel strength. After measuring the required thickening time with the consistometer another thickening time test is performed in the consistometer but the agitation of the slurry in the cell is ceased by stopping the motor after the estimated placement time and waiting for the estimated residence time in the loss zone. Then the motor is turned on again after waiting and the new viscosity is measured. The expected result is for the viscosity to go all the way to 100 Bearden units of consistency (BC) immediately after turning on the motor, which indicates a slurry that is too viscous to pump and has developed more than 250 lbs per 100 ft² of gel strength. See figure 6. The degree of thixotropy in the cement slurry can be adjusted with chemical additives such as bentonite, SMS, or gypsum.

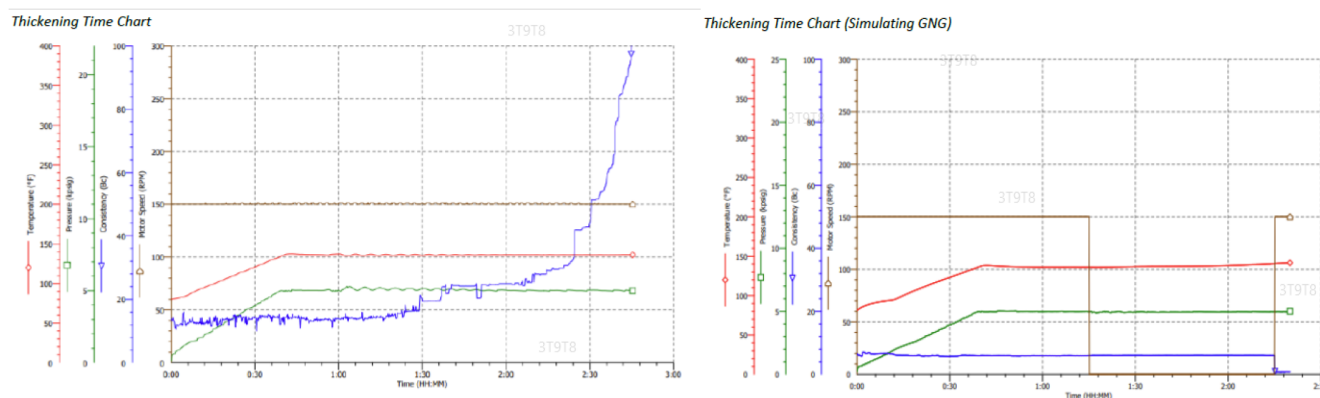


Figure 6—Thickening time test of a thixotropic cement slurry followed by a thickening time test of the same slurry with a shut down time. After the shut down time the pin was sheared immediately when the motor was activated and stopped the test.

The fluid train is pumped into the well through the drill pipe or tubing as a cement squeeze job using either the BOP method or with a down hole tool. If the operator is willing to take some risk, the fluid train could be pumped through the bottom hole assembly when the lost circulation is encountered, using the BOP cement squeeze method. Since these highly reactive and aggressive fluids are being pumped through the same pipe used for drilling the well, it is important to clean the pipe after the treatment. Foam wiper balls are pumped through the work string after circulating out excess cement. Also, the work string is flushed with a thick fluid containing cement retarder.

The work string for the cement squeeze is run to a depth above the lost circulation zone, if using the BOP squeeze method, or the downhole packer is set in the previous shoe or above the perforations if using

the downhole tool method. If using the BOP squeeze method, the work string can be run to the same depth as the losses or even below the losses. There are two schools of thought regarding the depth of the work string in relation to the depth of the losses. If the work string is at the same depth or close to the depth of the loss zone, then the fluids will be pumped directly into the loss zone, reducing the chances of the fluids becoming contaminated by well bore fluids before reaching the loss zone. The disadvantage of this method is the risk of getting the pipe stuck if the cement packs off or sets around the bottom of the drill pipe. For this reason, fluids can be pumped down the choke and kill line during the squeeze operation. If the end of the work string is placed several hundred feet above the loss zone then the risk of getting stuck is mitigated, however the fluids may not ever reach the loss zone before they are contaminated by well bore fluids, or the fluids react and plug the well bore above the loss zone. The latter case would regain circulation but if the cement is drilled out the losses would resume again as the loss zone is still not sealed. The same strategy must be considered when using the downhole tool, either place the tool right above the perforations or several hundred feet above.

Before the treatment, the operator would determine a maximum allowable surface pressure, in other words, the amount of pressure that would reflect an improvement in the losses. All of the pumping is done by the cement service company cement pumping unit. The fluid sequence would be fresh water spacer ahead, calcium chloride brine, fresh water spacer, sodium silicate solution, fresh water spacer, thixotropic cement, fresh water spacer and finally displacement fluid. If there is no evidence of curing the losses the entire sequence can be repeated again, even with a different cement slurry without waiting on cement. The calcium chloride and sodium silicate are in liquid form, prepared and stored in chemical tanks that are connected directly to the cement pump unit. It is advisable to use one cement pumping unit to pump the calcium chloride brine and sodium silicate solution, and another cement pump unit to pump the cement slurry to prevent any mixing of the reactive fluids in the mixing system.

During the displacement of the cement squeeze operation, the pumping rate can be reduced if there are any increases in pressure but stopping is not an option until the cement is pumped out of the work string. Any stoppage time could result in the cement slurry becoming too thick to pump and circulate out of the work string. The cement should be displaced at least 250 ft below the work string, unless the work string is near the loss zone or if the downhole tool is near the perforations. After the cement squeeze is displaced then the BOP is opened or the work string is disconnected from the packer. The work string is then picked up to a safe depth, preferably into the previous shoe, a foam wiper ball and cleaning fluid is pumped down the work string to remove any excess cement. After waiting on cement to set, the operator will attempt to establish circulation or determine if the losses have been reduced if unable to establish circulation. If there are signs that the losses have been reduced or pressure increases rapidly, locks up, then the pumping is stopped and the BOP is opened or the work string is disconnected from the downhole packer to circulate out any excess cement in the pipe and the work string is picked up to a safe depth.

Presentation of Data and Results

The novel lost circulation cement squeeze treatment has been applied on multiple wells since its first deployment. Three examples will be presented in this paper to demonstrate the effectiveness of the treatment. For each example there had been many attempts to cure the losses with conventional cement plugs and mud LCM pills. Only after these conventional LCM methods had failed to cure losses did the operator choose to use the new method. See [table 5](#) for a summary of all three examples.

Well A, Cement Squeeze to Cure Losses with A Cement Retainer

Well A encountered losses while drilling a surface hole section and finished drilling the section blind. The plan was to run surface casing with a stage tool and cement the casing in two stages. After cementing the casing in two stages and drilling out the stage tool plugs and cement, total losses were encountered,

suggesting that the casing was damaged at some depth. After determining the casing leak depth with a retrievable packer, it was decided to run a cement retainer above the casing leak and cure the losses with cement. Two attempts were made to cure the casing leak by squeezing conventional 15.8 ppg cement slurry below the retainer with no success. For the third attempt it was decided to use the new cement squeeze method with reactive fluids and thixotropic cement. See [table 2](#) below, which outlines the fluid sequence that was used.

Table 2—Fluid Sequence for Well A Lost Circulation Treatment Using A Downhole Tool, Cement Retainer

	Fluid Sequence	Volume (BBLS)	Density (PPG)	Rate (BPM)	Comment
1	Fresh Water Spacer	10	8.33	2-3	Spacer ahead
2	Calcium Chloride Brine	100	10.48	2-3	35 lbs/bbl Calcium Chloride
3	Fresh Water Spacer	10	8.33	2-3	Spacer between fluids
4	Sodium Silicate Solution	100	9.88	2-3	50:50 Sodium Silicate: Water
5	Fresh Water Spacer	10	8.33	2-3	Spacer between fluids
6	Cement Slurry 1	150	13.50	2-3	13.5 ppg Thixotropic Slurry
7	Fresh Water Spacer	10	8.33	2-3	Spacer ahead
8	Calcium Chloride Brine	100	10.48	2-3	35 lbs/bbl Calcium Chloride
9	Fresh Water Spacer	10	8.33	2-3	Spacer between fluids
10	Sodium Silicate Solution	100	9.88	2-3	50:50 Sodium Silicate: Water
11	Fresh Water Spacer	10	8.33	2-3	Spacer between fluids
12	Cement Slurry 2	150	15.80	2-3	15.8 ppg Thixotropic Slurry

While pumping the third cement squeeze with the new method the formation locked up and held 1000 psi. After drilling out the cement retainer and cleaning out cement across the casing leak depth, it was possible to maintain full returns and a successful positive pressure test. The next section was drilled, cased and cemented with full returns.

Well B, Cement Squeeze to Cure Losses Using the BOP Method

Well B encountered losses while drilling an intermediate hole section and it was not possible to finish drilling the section blind. There was a flow zone above the loss zone, which required a minimum amount of hydrostatic pressure to keep under control. Because of this the operator was forced to pump kill weight mud down the backside until the loss zone was cured. There were four attempts to cure the losses using water-based mud (WBM) loaded with LCM. After each attempt the losses remained unchanged so it was decided to use cement to cure the losses with the new cement squeeze method. Before attempting the cement squeeze the BHA was pulled out of the hole and a tubing stinger was run in the hole on drill pipe 200 ft above the loss zone. The tubing stinger was designed with a lower yield point, so if the cement were to set around the tubing the operator could break free from the tubing with over pull at surface.

After the work string was placed above the loss zone, the reactive fluid and cement slurry was pumped down the drill pipe while pumping mud down the backside. After four attempts using the new cement squeeze method the losses were cured and full circulation was regained. The rate of losses was progressively reduced after each placement. After the first squeeze, the rate of losses was reduced from 480 to 300 bbls per hour. After the second squeeze the rate of losses was reduced from 300 to 240 bbls per hour. After the third squeeze the rate of losses was reduced from 240 to 190 bbls per hour. After the fourth and last squeeze the rate of losses was reduced from 190 bbls per hour to full circulation. See [table 3](#) below, which outlines the fluid sequence that was used for each attempt. Well B is an interesting example, as it shows that the losses are reduced after each cement squeeze, which suggests that it may be possible to cure the losses in one operation if the volumes of reactive fluid and thixotropic cement are increased to match the loss rate.

Table 3—Fluid Sequence for Well B Lost Circulation Treatment Using the BOP Method

	Fluid Sequence	Volume (BBLS)	Density (PPG)	Rate (BPM)	Comment
1	Fresh Water Spacer	10	8.33	2-3	Spacer ahead
2	Sodium Silicate Solution	50	9.88	2-3	50:50 Sodium Silicate: Water
3	Fresh Water Spacer	10	8.33	2-3	Spacer between Fluids
4	Cement Slurry	105	13.6	2-3	13.6 ppg Thixotropic Slurry
5	Fresh Water Spacer	10	8.33	2-3	Spacer between Fluids
6	Displacement with WBM	117	10.5	2-3	Displacement Fluid

Well C, Cement Squeeze to Cure Losses Using the BOP Method

Well C encountered losses while drilling a production hole section and it was not possible to finish drilling the section blind. Since this was a production interval, there were flow zones that have to be isolated with cement after running the production long string. Similar to Well B, the operator was forced to pump kill weight mud down the backside until the loss zone was cured. Like Well B, there were four attempts to cure the losses using WBM loaded with LCM and after each attempt the losses remained unchanged. Also, a conventional balanced cement plug was attempted, but the losses remained unchanged. It was decided to use cement to cure the losses with the new cement squeeze method. Before attempting the cement squeeze the BHA was pulled out of the hole and a breakable tubing stinger was run in the hole on drill pipe 200 ft above the loss zone.

After three cement squeezes the losses were cured and full circulation was regained. Like Well B the rate of losses slowed down progressively after each squeeze. After the first squeeze, the rate of losses was reduced from total to 260 bbls per hour. After the second squeeze the rate of losses was reduced from 260 to 200 bbls per hour. After the third squeeze the rate of losses was reduced from 200 to full circulation. See [table 4](#) below, which outlines the fluid sequence that was used for each attempt. Well C is an interesting example as the operator also included dumping gravel into the loss zone before attempting to cure losses with cement, which actually temporarily cured the losses until the gravel was cleaned out. This suggests that a combination of cement and gravel could be an even more effective method to cure losses. Also, Well C illustrates that the volume of reactive fluids and thixotropic cement could have been increased to have a better chance of success on the first attempt.

Table 4—Fluid Sequence for Well C Lost Circulation Treatment Using the BOP Method

	Fluid Sequence	Volume (BBLS)	Density (PPG)	Rate (BPM)	Comment
1	Fresh Water Spacer	50	8.33	2-3	Spacer ahead
2	Calcium Chloride Brine	50	10.48	2-3	35 lbs/bbl Calcium Chloride
3	Fresh Water Spacer	10	8.33	2-3	Spacer between fluids
4	Sodium Silicate Solution	50	9.88	2-3	50:50 Sodium Silicate: Water
5	Fresh Water Spacer	10	8.33	2-3	Spacer between fluids
6	Cement Slurry	200	13.5	2-3	13.5 ppg Thixotropic Slurry
7	Fresh Water Spacer	30	8.33	2-3	Spacer between fluids
8	Displacement with WBM	134	11.4	2-3	Displacement Fluid

Table 5—Summary of three example wells and the order of LCM treatments applied to treat the loss zones

Well	Pre-Job Dynamic Losses (BBL/HR)	Pre-Job Static Losses (BBL/HR)	Description of Lost Circulation Treatments		Post-Job Dynamic Losses (BB/HR)	Post-Job Static Losses (BBL/HR)	Method of Squeeze
A	240	NA	1st	100 bbls of water as spacer ahead, 300 bbls of 15.8 ppg Conventional Cement Slurry.	240	NA	Down Hole Packer, Cement Retainer
	240	NA	2nd	100 bbls of water as spacer ahead, 500 bbls of 15.8 ppg Conventional Cement Slurry.	300	NA	
	300	NA	3rd	100 bbls of CaCl, 100 bbls of Sodium Silicate, 150 bbls of 13.5 ppg Thixotropic Cement, 100 bbls of CaCl 100 bbls of Sodium Silicate, 150 bbls of 15.8 ppg Thixotropic Cement Slurry.	Locked up, full circ. post drill out	NA	
B	200	150	1st	170 bbls of 10.5 ppg WBM with 286 PPB of LCM	Total	150	BOP Squeeze Method
	Total	150	2nd	200 bbls of 10.5 ppg WBM with 286 PPB of LCM	Total	150	
	Total	150	3rd	200 bbls of 10.5 ppg WBM with 286 PPB of LCM	Total	150	
	Total	150	4th	200 bbls of 10.5 ppg WBM with 286 PPB of LCM	480	150	
	480	150	5th	50 bbls of Sodium Silicate, 105 bbls of 13.6 ppg Thixotropic Cement Slurry	300	130	
	300	130	6th	50 bbls of Sodium Silicate, 105 bbls of 13.6 ppg Thixotropic Cement Slurry	240	100	
	240	100	7th	50 bbls of Sodium Silicate, 105 bbls of 13.6 ppg Thixotropic Cement Slurry	190	70	
	190	70	8th	50 bbls of Sodium Silicate, 105 bbls of 13.6 ppg Thixotropic Cement Slurry	Full Circ.	0	
C	Total	120	1st	150 bbls of 11.5 ppg WBM with 250 PPB of LCM	Total	120	BOP Squeeze Method
	Total	120	2nd	250 bbls of 11.5 ppg WBM with 250 PPB of LCM	Total	120	
	Total	120	3rd	200 bbls of 11.5 ppg WBM with 250 PPB of LCM	Total	60	
	Total	60	4th	250 bbls of 11.5 ppg WBM with 250 PPB of LCM	Total	60	
	Total	60	5th	Dumped Gravel down the annulus	168	0	
	168	0	6th	100 bbls of 14.9 ppg Conventional Cement Slurry	Total	60	
	Total	60	7th	50 bbls of CaCl, 50 bbls of Sodium Silicate, 200 bbls of 14.9 ppg Thixotropic Cement Slurry	260	60	
	260	60	8th	50 bbls of CaCl, 30 bbls of Sodium Silicate, 100 bbls of 14.9 ppg Thixotropic Cement Slurry	200	60	
	200	60	9th	50 bbls of CaCl, 50 bbls of Sodium Silicate, 300 bbls of 14.9 ppg Thixotropic Cement Slurry	Full Circ.	0	

Conclusions

The novel lost circulation cement squeeze treatment has proven to be an effective method for curing severe to total losses in drilling operations. The combination of reactive fluids, such as calcium chloride brine and sodium silicate solution, with thixotropic cement slurries has shown to rapidly seal off loss zones and

restore well integrity. Although these methods are old and well documented in previous works, they are still simple and effective techniques to curing lost circulation. For severe to complete lost circulation, before attempting to cure the losses with a balanced cement plug, it is much more efficient to begin with this novel lost circulation treatment. The examples presented in this paper demonstrate the success of this method in various well scenarios, highlighting its potential to reduce non-productive time and operational costs. Further research and development of this technology may lead to even more efficient and effective lost circulation treatments, ultimately improving the overall efficiency and safety of drilling operations

Nomenclature

BBS	-	Barrels
BC	-	Bearden Units of Consistency
BPH	-	Barrels Per Hour
BOP	-	Blowout Preventer
BPM	-	Barrels Per Minute
BWOC	-	By Weight of Cement
C-S-H	-	Calcium Silicate Hydrate
GPS	-	Gallons Per Sack of Cement
LBS	-	Pounds
LCM	-	Lost Circulation Material
OBM	-	Oil Based Mud
PPB	-	Pounds Per Barrel
PPG	-	Pounds Per Gallon
SMS	-	Sodium Metasilicate
ty	-	Yield Point
up	-	Viscosity, Centipoise
WBM	-	Water Based Mud

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